221-2327 A#03 M.Rohaan Sec-E

Q#01:

Sol.

a) $\sum P(X=x) = 1$

$c + c^2 + c^2 + c + 3c^2 + 2c = 1$

$5c^2 + 4c = 1$

$5c^2 + 4c - 1 = 0$

$(5c - 1)(c + 1) = 0$

$c = \frac{1}{5}$ or $c = -1$ ∵ Probability can't be negative

$$\boxed{c = \frac{1}{5}}$$

b)

x	0	1	2	3
$P(x)$	$\frac{1}{5}$	$\frac{1}{25}$	$\frac{6}{25}$	$\frac{13}{25}$

$F(x) = P(X \leq x)$

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{5}, & 0 \leq x < 1 \\ \frac{1}{5} + \frac{1}{25}, & 1 \leq x < 2 \\ \frac{1}{5} + \frac{1}{25} + \frac{6}{25}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

∵ Because Total Probability has full

$$F(x)=\begin{cases}0, & x<0\\ 1/5, & 0\le x<1\\ 6/25, & 1\le x<2\\ 12/25, & 2\le x<3\\ 1, & x\ge 3\end{cases}$$

c) $E(X)=\sum x\cdot P(X=x)$

$= 0\cdot\frac{1}{5}+1\cdot\frac{1}{25}+2\cdot\frac{6}{25}+3\cdot\frac{13}{25}$

$= 52/25$

d) $V(X)=E(X^2)-(E(X))^2$

$E(X^2)=\sum x^2\cdot P(X=x)$

$= 0^2\cdot\frac{1}{5}+1^2\cdot\frac{1}{25}+2^2\cdot\frac{6}{25}+3^2\cdot\frac{13}{25}$

$= 142/25$

$V(X)=\frac{142}{25}-\left(\frac{52}{25}\right)^2=\frac{846}{625}$

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e) $E(2X+3) = 2E(X)+3 = 2.\frac{52}{25}+3 = \frac{179}{25}$

$V(5X+4) = 5^2 . V(X) = 25.\frac{846}{625} = \frac{846}{25}$

Q#02:-

Sol.

a) $\{1, 2, 3, \ldots, 100\}$

X = Remainder when the number is divided by 7

Possible remainders are = 0, 1, 2, 3, 4, 5, 6.

Note That, Maximum number before 100 is 98 which gives remainder 0 when divide by 7.

$$7 \overline{)\,98}\;\;14,\quad -98,\quad 0$$

Means each remainder repeats 14 times but after 98, Two numbers 99 and 100 when divide remainder 1 and 2 repeats one extra time. $15+15+14+14+14+14+14 = 100$

Remainder 1 appears — 15 times

Remainder 2 appears — 15 times

Remainder 0, 3, 4, 5, 6 appears — 14 times

$P(X=0) = 14/100$

$P(X=1) = 15/100$

$P(X=2) = 15/100$

$P(X=3) = 14/100$

$P(X=4) = 14/100$

$P(X=5) = 14/100$

$P(X=6) = 14/100$

x	0	1	2	3	4	5	6
$P(X=x)$	$\frac{14}{100}$	$\frac{15}{100}$	$\frac{15}{100}$	$\frac{14}{100}$	$\frac{14}{100}$	$\frac{14}{100}$	$\frac{14}{100}$

b) $P(1 \leq X < 5)$

$= P(1) + P(2) + P(3) + P(4) + P(5)$

$= \frac{15}{100} + \frac{15}{100} + \frac{14}{100} + \frac{14}{100} + \frac{14}{100} = \frac{18}{25}$

c) $P(X \leq 3)$

$= P(0) + P(1) + P(2) + P(3)$

$= \frac{14}{100} + \frac{15}{100} + \frac{15}{100} + \frac{14}{100} = \frac{29}{50}$

d) $P(X \geq 4)$

$= P(4) + P(5) + P(6)$

$= \frac{14}{100} + \frac{14}{100} + \frac{14}{100} = \frac{21}{50}$

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where X can be $x = -1, 0, 1$

$|-2| = 2 \leq 1$ X

$|-1| = 1 \leq 1$ ✓ $\quad |2| = 2 \leq 1$ X

e) $P(|X| \leq 1)$

$= P(0) + P(1) = \frac{14}{100} + \frac{15}{100} = \frac{29}{100}$

Q#03:-

Sol.

a) $P(0.5 \leq X \leq 1.5)$

To find k, $\int_{-\infty}^{\infty} f(x) = 1$ ∵ Total Probability is always 1

$$\int_0^1 (kx)\,dx + \int_1^2 (k(2-x))\,dx = 1$$

$$k\left|\frac{x^2}{2}\right|_0^1 + k\left|2x - \frac{x^2}{2}\right|_1^2 = 1$$

$$k\left[\frac{1^2}{2} - \frac{0^2}{2}\right] + k\left[2(2) - \frac{2^2}{2} - \left(2(1) - \frac{1^2}{2}\right)\right] = 1$$

$$k\left(\frac{1}{2}\right) + k\left(\frac{1}{2}\right) = 1$$

$$\boxed{k = 1}$$

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$P(0.5 \le X \le 1.5) = \int_{0.5}^{1} x\,dx + \int_{1}^{1.5} (2-x)\,dx$$

$$= \left| \frac{x^2}{2} \right|_{0.5}^{1} + \left| 2x - \frac{x^2}{2} \right|_{1}^{1.5}$$

$$= \left[\frac{1^2}{2} - \frac{0.5^2}{2} \right] + \left[2(1.5) - \frac{1.5^2}{2} - \left(2(1) - \frac{1^2}{2} \right) \right]$$

$$= \frac{3}{4}$$

b) $$E(X) = \int_{-\infty}^{+\infty} x \cdot f(x)\,dx$$

$$= \int_{0}^{1} (x \cdot x)\,dx + \int_{1}^{2} (x(2-x))\,dx = \int_{0}^{1} x^2\,dx + \int_{1}^{2} (2x - x^2)\,dx$$

$$= \left| \frac{x^3}{3} \right|_{0}^{1} + \left| \frac{2x^2}{2} - \frac{x^3}{3} \right|_{1}^{2}$$

$$= \left[\frac{1^3}{3} - \frac{0^3}{3} \right] + \left[2^2 - \frac{2^3}{3} - \left(1^2 - \frac{1^3}{3} \right) \right] = 1$$

c) $$V(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \int_{-\infty}^{+\infty} x^2 \cdot f(x)\,dx = \int_{0}^{1} (x^2 \cdot x)\,dx + \int_{1}^{2} (x^2(2-x))\,dx$$

$$= \int_0^1 x^3\,dx + \int_1^2 (2x^2 - x^3)\,dx$$

$$= \left|\frac{x^4}{4}\right|_0^1 + \left|\frac{2x^3}{3} - \frac{x^4}{4}\right|_1^2$$

$$= \left[\frac{1^4}{4} - \frac{0^4}{4}\right] + \left[\frac{2(2^3)}{3} - \frac{2^4}{4} - \left(\frac{2(1^3)}{3} - \frac{1^4}{4}\right)\right]$$

$$= \frac{7}{6}$$

$$V(X) = \frac{7}{6} - 1^2 = \frac{1}{6}$$

Q#04:-

Sol.

$0 < x < 2$

$\because (2-x)(2+x) = 4 - x^2 \Rightarrow f(x) = A(4-x^2)$

a) $\int_0^2 A(4-x^2)\,dx = 1$ $\because$ Total Probability $= 1$

$$A\left[4x - \frac{x^3}{3}\right]_0^2 = 1$$

$$A\left[4(2) - \frac{2^3}{3} - \left(4(0) - \frac{0^3}{3}\right)\right] = 1$$

$$A \cdot \frac{16}{3} = 1$$

$$\boxed{A = \frac{3}{16}}$$

$$f(x) = \frac{3}{16}(4 - x^2), \quad 0 < x < 2$$

b) $$P(X \leq 1) = \int_0^1 \left(\frac{3}{16}(4 - x^2)\right) dx$$

$$\frac{3}{16}\left[4x - \frac{x^3}{3}\right]_0^1 = \frac{3}{16}\left[4(1) - \frac{1^3}{3} - \left(4(0) - \frac{0^3}{3}\right)\right]$$

$$P(X \leq 1) = 11/16$$

c) $$P(X \geq 2) = \int_2^2 \left(\frac{3}{16}(4 - x^2)\right) dx$$

Since X is continuous and PDF is only for $0 < x < 2$, otherwise zero, So So $P(X \geq 2) = 0$, Because for CRV, probability at a single point is zero

d) $P(1<X<2) = \int_1^2 \left(\frac{3}{16}(4-x^2)\right) dx$

$= \frac{3}{16}\left[4x - \frac{x^3}{3}\right]_1^2 = \frac{3}{16}\left[4(2) - \frac{2^3}{3} - \left(4(1) - \frac{1^3}{3}\right)\right]$

$= \frac{5}{16}$

e) Median $\rightarrow P(X \leq m) = 0.5$

$$\int_0^m \left(\frac{3}{16}(4-x^2)\right) dx = 0.5$$

$$\frac{3}{16}\left[4x - \frac{x^3}{3}\right]_0^m = 0.5$$

$$\frac{3}{16}\left[4m - \frac{m^3}{3}\right] = 0.5$$

$m^3 - 12m + 8 = 0$

$m_1 = -3.758$, $m_2 = 3.064$, $m_3 = 0.6945$

$m_1 \Rightarrow$ negative and out of interval $0 \leq x \leq 2$

$m_2 \Rightarrow$ out of interval $0 \leq x \leq 2$

m_3 ✓

median = 0.695

Mode → Global Maxima

$f(x)=\frac{3}{16}(4-x^2)$ $[0,2]$

$f'(x)=-2x$, $f'(x)=0$

$-2x=0 \Rightarrow \boxed{x=0}$

Put $x=0$, $x=2$ in $f(x)$

$f(0)=\frac{12}{16}=\frac{3}{4}$ ✓

$f(2)=0$

$\boxed{\text{mode}=0}$

Q#05:-

Sol.

a) Total Probability = 1

$p+p+0.1+q+q=1$

$p+q=0.45$ —— (i)

$P(X\geq 0)=3P(X<0)$

$P(X\geq 0)=P(0)+P(1)+P(2)=0.1+q+q=0.1+2q$

$P(X<0)=P(-2)+P(-1)=p+p=2p$

$0.1+2q=3(2p)$

$0.1 + 2q = 6p$ — (ii)

from (i)

$q = 0.45 - p$ — (iii)

Put q in (ii)

$0.1 + 2(0.45 - p) = 6p$

$\boxed{p = \frac{1}{8}}$ Put p in (iii)

$q = 0.45 - \frac{1}{8} \boxed{= \frac{13}{40}}$

b) $F(x) = P(X \leq x)$

$$F(x) = \begin{cases} 0, & x < -2 \\ \frac{1}{8}, & -2 \leq x < -1 \\ \frac{1}{8} + \frac{1}{8}, & -1 \leq x < 0 \\ \frac{1}{8} + \frac{1}{8} + \frac{1}{10}, & 0 \leq x < 1 \\ \frac{1}{8} + \frac{1}{8} + \frac{1}{10} + \frac{13}{40}, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

$$F(x) = \begin{cases} 0, & x < -2 \\ \frac{1}{8}, & -2 \leq x < -1 \\ \frac{2}{8}, & -1 \leq x < 0 \\ 7/20, & 0 \leq x < 1 \\ 27/40, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

$P(X \leq 1) = F(1) = \frac{27}{40}$

c) $E(X) = \sum x P(X=x)$

$= (-2)\left(\frac{1}{8}\right) + (-1)\left(\frac{1}{8}\right) + 0\left(\frac{1}{10}\right) + 1\left(\frac{13}{40}\right) + 2\left(\frac{13}{40}\right)$

$= \frac{3}{5}$

d) $V(X) = E(X^2) - (E(X))^2$

$E(X^2) = \sum x^2 \cdot P(x) = (-2)^2\left(\frac{1}{8}\right) + (-1)^2\left(\frac{1}{8}\right) + (0^2)\left(\frac{1}{10}\right) + (1)^2\left(\frac{13}{40}\right) + (2)^2\left(\frac{13}{40}\right) = \frac{9}{4}$

$V(X) = \frac{9}{4} - \left(\frac{3}{5}\right)^2 = \frac{189}{100}$

e) $E(5X+3) = 5(E(X)) + 3 = 5\left(\frac{3}{5}\right) + 3 = 6$

$V(2X+1) = 2^2(V(X)) = 4 \cdot \frac{189}{100} = \frac{189}{25}$

Q#06:-

Sol.

a) Marginal Distribution of X

$P(X=1) = 0.05 + 0.05 + 0.00 = 0.10$

$P(X=2) = 0.05 + 0.10 + 0.20 = 0.35$

$P(X=3) = 0.10 + 0.35 + 0.10 = 0.55$

Marginal Distribution of Y

$P(Y=1) = 0.05 + 0.05 + 0.10 = 0.20$

$P(Y=2) = 0.05 + 0.10 + 0.35 = 0.50$

$P(Y=3) = 0.00 + 0.20 + 0.10 = 0.30$

b) $P(Y=3 \mid X=2) = \frac{P(X=2, Y=3)}{P(X=2)} = \frac{0.10}{0.35} = \frac{2}{7}$

$P(X \geq Y) = 0.05 + 0.10 = 0.15$

$P(X+Y \leq 3) = 0.05 + 0.05 = 0.10$

c) $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$ for all values

$P(X=1, Y=1) = 0.05$

But $P(X=1) \cdot P(Y=1) = 0.10 \cdot 0.20 = 0.02$

$0.05 \neq 0.02$

X and Y are not independent

d) $E(X+Y) = E(X) + E(Y)$

$E(X) = (1)(0.10) + (2)(0.35) + (3)(0.55) = 2.45$

$E(Y) = (1)(0.20) + (3)(0.50) + (5)(0.30) = 3.20$

$E(X+Y) = 2.45 + 3.20 = 5.65$

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* $E(X-Y) = E(X) - E(Y) = 2.45 - 3.20 = -0.75$

* $E(XY) = \sum\sum xy\, f(x,y)$

$y=1 \Rightarrow (1)(1)(0.05) + (2)(1)(0.05) + (3)(1)(0.10) = 0.45$

$y=3 \Rightarrow (1)(3)(0.05) + (2)(3)(0.10) + (3)(3)(0.35) = 3.90$

$y=5 \Rightarrow (1)(5)(0.00) + (2)(5)(0.20) + (3)(5)(0.10) = 3.50$

$E(XY) = 0.45 + 3.90 + 3.50 = 7.85$

e) $COV(X,Y) = E(XY) - E(X)\cdot E(Y)$

$= 7.85 - (2.45)(3.20) = 0.01$

$$CORR(X,Y) = \frac{COV(X,Y)}{\sqrt{V(X)}\cdot\sqrt{V(Y)}}$$

$V(X) = E(X^2) - (E(X))^2$

$E(X^2) = 1^2(0.10) + 2^2(0.35) + 3^2(0.55) = 6.45$

$V(X) = 6.45 - (2.45)^2 = 0.4475$

$V(Y) = E(Y^2) - (E(Y))^2$

$$E(Y^2) = 1^2(0.20) + 3^2(0.50) + 5^2(0.30) = 12.20$$

$$V(Y) = 12.20 - (3.20)^2 = 1.96$$

~~Cov(X,Y) of~~ $Corr(X,Y) = \dfrac{0.01}{\sqrt{0.669} \cdot \sqrt{1.4}} = 0.011$

Q#07:-

Sol.

a) X = number of oranges, Y = number of apples

Then Bananas in Sample = $4 - X - Y$

Total ways to choose 4 fruits from 8

$$\binom{8}{4} = 70$$

So, every Probability

$$P(X=x, Y=y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{4-x-y}}{\binom{8}{4}}$$

where $4-x-y$ must between 0 and 3.

we find all possible (x, y)

* $0 \le x \le 3$ * $0 \le y \le 2$ * $0 \le 4-x-y \le 3$

1) $y=0$, Then bananas $= 4-x$, $x=1,2,3$

(1,0)

$$P(1,0) = \frac{\binom{3}{1}\binom{2}{0}\binom{3}{3}}{70} = \frac{3}{70}$$

(2,0)

$$P(2,0) = \frac{\binom{3}{2}\binom{2}{0}\binom{3}{2}}{70} = \frac{9}{70}$$

(3,0)

$$P(3,0) = \frac{\binom{3}{3}\binom{2}{0}\binom{3}{1}}{70} = \frac{3}{70}$$

2) $y=1$, Then bananas $= 3-x$, $x=0,1,2,3$

(0,1)

$$P(0,1) = \frac{\binom{3}{0}\binom{2}{1}\binom{3}{3}}{70} = 2/70$$

(1,1)

$$P(1,1) = \frac{\binom{3}{1}\binom{2}{1}\binom{3}{2}}{70} = 18/70$$

(2,1)

$$P(2,1) = \frac{\binom{3}{2}\binom{2}{1}\binom{3}{1}}{70} = 18/70$$

(3,1)

$$P(3,1) = \frac{\binom{3}{3}\binom{2}{1}\binom{3}{0}}{70} = 2/70$$

3) $y=2$, Then bananas $= 2-x$, $x=0,1,2$

(0,2)

$$P(0,2) = \frac{\binom{3}{0}\binom{2}{2}\binom{3}{2}}{70} = 3/70$$

$$P(1,2) = \frac{\binom{3}{1}\binom{2}{2}\binom{3}{1}}{70} = 9/70$$

$$P(2,2) = \frac{\binom{3}{2}\binom{2}{2}\binom{3}{0}}{70} = 3/70$$

	$y=0$	$y=1$	$y=2$
$x=0$	0	$2/70$	$3/70$
$x=1$	$3/70$	$18/70$	$9/70$
$x=2$	$9/70$	$18/70$	$3/70$
$x=3$	$3/70$	$2/70$	0

$$\sum P(x,y) = \frac{70}{70} = 1$$

b) $$P(x+y \leq 2) = \frac{2+3+3+18+9}{70} = \frac{1}{2}$$

c) $$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

$P(X=0) = 5/70$	$P(Y=0) = 15/70$
$P(1) = 30/70$	$P(1) = 40/70$
$P(2) = 30/70$	$P(2) = 15/70$
$P(3) = 5/70$	

Now test $(x,y) = (0,1)$

$P(0,1) = 2/70$

but $P(X=0)P(Y=1) = \frac{5}{70} \cdot \frac{40}{70} = \frac{2}{49}$

$2/70 \neq 2/49$ → So X and Y are dependent

d) $E(X) = \sum x \cdot P(X=x)$

$= 0\cdot\frac{5}{70} + 1\cdot\frac{30}{70} + 2\cdot\frac{30}{70} + 3\cdot\frac{5}{70} = \frac{3}{2}$

$E(Y) = \sum y \cdot P(Y=y)$

$= 0\cdot\frac{15}{70} + 1\cdot\frac{40}{70} + 2\cdot\frac{15}{70} = 1$

e) $Cov(X,Y) = E(XY) - E(X)E(Y)$

$E(XY) = \sum xy\, P(x,y)$

$= (1)(1)\left(\frac{18}{70}\right) + (1)(2)\left(\frac{9}{70}\right) + (2)(1)\left(\frac{18}{70}\right) + (2)(2)\left(\frac{3}{70}\right) + (3)(1)\left(\frac{2}{70}\right)$

$= 9/7$

$Cov(X,Y) = \frac{9}{7} - \left(\frac{3}{2}\right)(1) = -\frac{3}{14}$

$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X)}\cdot\sqrt{V(Y)}}$

$V(X) = E(X^2) - E(X)$

$E(X^2) = 0^2\cdot\frac{5}{70} + 1^2\cdot\frac{30}{70} + 2^2\cdot\frac{30}{70} + 3^2\cdot\frac{5}{70} = \frac{39}{14}$

$V(X) = \frac{39}{14} - \left(\frac{3}{2}\right)^2 = \frac{15}{28}$

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$$V(Y) = E(Y^2) - (E(Y))^2$$

$$E(Y^2) = 0^2 \cdot \frac{15}{70} + 1^2 \cdot \frac{40}{70} + 2^2 \cdot \frac{15}{70} = \frac{10}{7}$$

$$V(Y) = \frac{10}{7} - (1)^2 = \frac{3}{7}$$

$$Corr(X,Y) = \frac{-3/14}{\sqrt{\frac{15}{28}} \cdot \sqrt{\frac{3}{7}}} \approx -0.447$$

Q#08:

Sol.

a) Marginal of X

$$f_X(x) = \int_1^2 \frac{3x - y}{9}\, dy$$

$$= \frac{1}{9}\left[3xy - \frac{y^2}{2}\right]_1^2$$

$$= \frac{1}{9}\left((6x - 2) - \left(3x - \frac{1}{2}\right)\right) = \frac{2x-1}{6}$$

$$f_X(x) = \begin{cases} \frac{2x-1}{6}, & 1 \leq x < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_1^3 \frac{3x-y}{9}\,dx$$

$$= \frac{1}{9}\left[\frac{3x^2}{2} - xy\right]_1^3$$

$$= \frac{1}{9}\left(\frac{27}{2} - 3y - \left(\frac{3}{2} - y\right)\right) = \frac{1}{9}(12-2y)$$

$$\therefore f_Y(y) = \begin{cases} \frac{12-2y}{9}, & 1<y<2 \\ 0, & \text{otherwise} \end{cases}$$

b) $P(1.5<X<2.5, 0.5<Y<1.5)$

$$= \int_{1.5}^{2.5}\int_1^{1.5}\left(\frac{3x-y}{9}\right)dy\,dx$$

$$\int_1^{1.5}\left(\frac{3x-y}{9}\right)dy = \frac{1}{9}\left[3xy - \frac{y^2}{2}\right]_1^{1.5}$$

$$= \frac{1}{9}\left[3(x)(1.5) - \frac{(1.5)^2}{2} - \left(3(x)(1) - \frac{1^2}{2}\right)\right]$$

$$= \frac{x}{6} - \frac{5}{72}$$

$$\int_{1.5}^{2.5} \left(\frac{x}{6} - \frac{5}{72}\right) dx = \frac{19}{72}$$

$$P(X>2) = \int_2^3 f_X(x)\,dx = \int_2^3 \frac{2x-1}{6}\,dx$$

$$= \left[\frac{x^2}{6} - \frac{x}{6}\right]_2^3 = \frac{2}{3}$$

$$P(Y \le 1.5) = \int_1^{1.5} f_Y(y)\,dy = \int_1^{1.5} \frac{12-2y}{9}\,dy$$

$$= \frac{1}{9}\left[12y - 2y^2\right]_1^{1.5} = \frac{19}{36}$$

c) $$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{(3x-y)/9}{(12-2y)/9}$$

$$= \frac{3x-y}{12-2y}$$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{(3x-y)/9}{(2x-1)/6} = \frac{2(3x-y)}{3(2x-1)}$$

for Independena,

$f(x,y) = f_X(x)\, f_Y(y)$

But here

$$f(x,y) = \frac{3x-y}{9}$$

and $f_X(x)\, f_Y(y) = \frac{2x-1}{6} \cdot \frac{12-2y}{9}$

Not equal, So

X and Y are dependent.

d) $P(X>Y)$, $1<y<2$; $y<x<3$

$$P(X>Y) = \int_1^2 \int_y^3 \left(\frac{3x-y}{9}\right) dx\, dy$$

$$\int_1^2 \frac{1}{9}\left[\frac{3x^2}{2} - xy\right]_y^3 = \int_1^2 \frac{1}{9}\left[\frac{27}{2} - 3y - \frac{3y^2}{2} + y^2\right]$$

$= 47/54$

$P(X+Y<1)$ $x>1$, $y>1$, $x+y>2$

$P(X+Y<1) = 0$

e) $Cov(X,Y) = E(XY) - E(X)E(Y)$

$$E(X) = \int_1^3 x f_X(x)\,dx = \int_1^3 x \cdot \frac{2x-1}{8}\,dx = \frac{1}{8}\left[\frac{2x^3}{3} - x\right]_1^3$$

$$= 20/9$$

$$E(Y) = \int_1^2 y \cdot f_Y(y)\,dy = \int_1^2 y \cdot \frac{12-2y}{9}\,dy = \frac{1}{9}\left[\frac{12y^2}{2} - \frac{2}{3}\right]$$

$$= 40/27$$

$$E(XY) = \int_1^3 \int_1^2 xy\,\frac{3x-y}{9}\,dy\,dx = \frac{89}{27}$$

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$

$$= \frac{89}{27} - \frac{20}{9} \cdot \frac{40}{27} = \frac{1}{243}$$

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X)} \cdot \sqrt{V(Y)}}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \int_1^3 x^2 \cdot f_X(x)\,dx = \int_1^3 x^2 \cdot \frac{2x-1}{8} = \frac{1}{8}\left[\frac{2x^4}{4} - \frac{x^3}{3}\right.$$

$$= 47/9$$

$$V(X) = \frac{47}{9} - \left(\frac{20}{9}\right)^2 = \frac{23}{81}$$

$$V(Y) = E(X^2) - (E(Y))^2$$

$$= \int_1^2 y^2 \cdot f_Y(y)\, dy = \int_1^2 y^2 \cdot \frac{12-2y}{9}\, dy$$

$$= \frac{1}{9}\left[\frac{12y^3}{3} - \frac{2y^4}{4}\right]_1^2 = \frac{41}{18}$$

$$V(Y) = \frac{41}{18} - \left(\frac{40}{27}\right)^2 = \frac{121}{1458}$$

$$Corr(X,Y) = \frac{1/243}{\sqrt{23/81} \cdot \sqrt{121/1458}} \approx 0.027$$

Q#09:-

Sol.

a) Marginal of $X \rightarrow f_X(x) = \int_0^1 24x^2y(1-x)\, dy$

$$= 24x^2(1-x)\int_0^1 y\, dy = 24x^2(1-x)\left[\frac{y^2}{2}\right]_0^1$$

$$= 24x^2(1-x)\cdot\frac{1}{2} = 12x^2(1-x)$$

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$$f_X(x) = \begin{cases} 12x^2(1-x), & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Marginal of $Y \rightarrow f_Y(y) = \int_0^1 24x^2y(1-x)\,dx$

$$= 24y \int_0^1 x^2(1-x)\,dx = 24y \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1$$

$$= 24y\left(\frac{1}{3} - \frac{1}{4}\right) = 2y$$

$$f_Y(y) = \begin{cases} 2y, & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

b) $$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{24x^2y(1-x)}{12x^2(1-x)}$$

$$= 2y$$

c) independent if $f(x,y) = f(x) \cdot f(y)$

$$f(x) \cdot f(y) = (12x(1-x))(2y) = 24x^2y(1-x)$$

X and Y are independent.

d) $E(XY) = E(X)E(Y)$

$$E(X) = \int_0^1 x \cdot f(x)\,dx = \int_0^1 x \cdot 12x^2(1-x)\,dx$$

$$= 12\int_0^1 (x^3 - x^4)\,dx = 12\left[\frac{x^4}{4} - \frac{x^5}{5}\right]_0^1 = \frac{3}{5}$$

$$E(Y) = \int_0^1 y \cdot f(y)\,dy = \int_0^1 y \cdot (2y)\,dy$$

$$= \left[2\,\frac{y^3}{3}\right]_0^1 = \frac{2}{3}$$

$$E(XY) = \frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}$$

e) $Cov(X,Y) = E(XY) - E(X)E(Y)$

$$= \frac{2}{5} - \frac{2}{5} = 0$$

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X)} \cdot \sqrt{V(Y)}} = 0$$

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Q#10:-

Sol.

a) $f(x,y) = f(x) \cdot f(y)$ for independent

$$f(x,y) = \left(\frac{3}{8}x^2\right)\left(\frac{3}{8}y^2\right) = \frac{9}{64}x^2y^2$$

for $0 \le x \le 2$, $0 \le y \le 2$

$$\begin{cases} \frac{9}{64}x^2y^2, & 0 \le x \le 2, 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

b) $P(X > Y)$

As X and Y are independent

$P(X > Y) = P(Y > X)$, $P(X = Y) = 0$

$$P(X > Y) + P(Y > X) = 1$$

$$2P(X > Y) = 1$$

$$\boxed{P(X > Y) = 1/2}$$

c) $P(X+Y \le 1) = \iint_{x+y\le 1} f(x,y)\,dx\,dy$

$0 \le x \le 1,\ 0 \le y \le 1-x$

$P(X+Y \le 1) = \int_0^1 \int_0^{1-x} \frac{9}{64} x^2 y^2 \,dy\,dx$

$\frac{9}{64} \int_0^1 x^2 \left(\int_0^{1-x} y^2 dy \right) dx =$

$\int_0^{1-x} y^2 dy = \frac{(1-x)^3}{3}$

$P(X+Y \le 1) = \frac{9}{64} \int_0^1 x^2 \frac{(1-x)^3}{3} dx$

$= \frac{3}{64} \int_0^1 x^2 (1-x)^3 dx = \frac{3}{64} \int_0^1 (x^2 - 3x^3 + 3x^4 - x^5) dx$

$= \left[\frac{x^3}{3} - \frac{3x^4}{4} + \frac{3x^5}{5} - \frac{x^6}{6} \right]_0^1 = \frac{1}{1280}$